



James Ruse Agricultural High School

2022 Year 12 TASK 3

Mathematics Extension 1

General

Instructions

- Reading time – 5 minutes
- Working time – 70 minutes
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/ or calculations

Total marks:
55

Section I – 5 marks (pages 2–3)

- Attempt Questions 1–5
- Allow about 7 minutes for this section

Section II – 50 marks (pages 4–10)

- Attempt Questions 6–12
- Allow about 1 hour and 3 minutes for this section

Section I

5 marks

Attempt Questions 1 – 5

Allow about 7 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 5.

- 1 The unit vector in the direction of $\begin{pmatrix} 7 \\ -3 \end{pmatrix}$ is
- A. $\frac{1}{40} \begin{pmatrix} 7 \\ -3 \end{pmatrix}$
- B. $\frac{1}{\sqrt{40}} \begin{pmatrix} 7 \\ -3 \end{pmatrix}$
- C. $\frac{1}{58} \begin{pmatrix} 7 \\ -3 \end{pmatrix}$
- D. $\frac{1}{\sqrt{58}} \begin{pmatrix} 7 \\ -3 \end{pmatrix}$
- 2 Suppose $\mathbf{x} \cdot \mathbf{y} = 2$, $|\mathbf{x}| = 2$ and $|\mathbf{y}| = \sqrt{2}$. The angle between the vectors $\mathbf{x}$ and $\mathbf{y}$ is
- A. 30°
- B. 45°
- C. 60°
- D. 90°
- 3 The coefficient of $a^{-6}b^4$ in the expansion of $\left(\frac{1}{a} - \frac{2b}{3}\right)^{10}$ is
- A. $\frac{1102}{27}$
- B. $\frac{1120}{72}$
- C. $\frac{1120}{27}$
- D. $\frac{1120}{37}$

4 Consider the statements below:

- I. If $|\underline{u}| \geq |\underline{v}|$ then $|\text{proj}_{\underline{v}}(\underline{u})| \geq |\text{proj}_{\underline{u}}(\underline{v})|$.
- II. The angle between $\underline{u}$ and $\underline{v}$ is the same as the angle between $\text{proj}_{\underline{v}}(\underline{u})$ and $\text{proj}_{\underline{u}}(\underline{v})$.

Which of these expressions are true for all two dimensional vectors $\underline{u}$ and $\underline{v}$?

- A. Neither I nor II are true.
- B. Only I is true.
- C. Only II is true.
- D. Both I and II are true.

5 Suppose the vectors $\mathbf{u}$ and $\mathbf{v}$ form any two sides of a regular hexagon. How many possible values of $\mathbf{u} \cdot \mathbf{v}$ are there?

- A. 2
- B. 3
- C. 4
- D. 6

Section II

50 marks

Attempt Questions 6–12

Allow about 1 hour and 3 minutes for this section

Answer each question on a new page. Extra writing pages are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (7 marks) Start a new page

- (a) Consider the points $P_1(3, 4)$ to $P_2(5, -1)$.
- (i) Write down the position vector of point P_1 . Give your answer as a column vector. **1**
 - (ii) Find the vector $\overrightarrow{P_1P_2}$. Give your answer in component form. **2**
 - (iii) Find the value of $|\overrightarrow{P_1P_2}|$. **1**
 - (iv) If $\overrightarrow{OP_1} = \mathbf{u}$ and $\overrightarrow{OP_2} = \mathbf{v}$ find $\overrightarrow{P_1P_2}$ in terms of $\mathbf{u}$ and $\mathbf{v}$. **1**
 - (v) Draw a diagram showing the vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{u} + \mathbf{v}$. **1**
 - (vi) Find the vector from the point A to the origin where $\overrightarrow{AP_2}$ is $2\mathbf{i} - 3\mathbf{j}$. **1**

End of Question 6

Question 7 (7 marks) [Start a new page](#)

- (a) A ship S is moving along a straight line with constant velocity in kilometres per hour.

When $t = 0$ hours the position vector is $\mathbf{s} = 9\mathbf{i} - 6\mathbf{j}$.

When $t = 4$ hours, the position vector is $\mathbf{s} = 21\mathbf{i} + 10\mathbf{j}$.

(i) Find the average speed of S . **2**

(ii) Find the direction in which S is moving, giving your answer as a bearing. **1**

(iii) Show that $\mathbf{s} = (3t + 9)\mathbf{i} + (4t - 6)\mathbf{j}$. **2**

- (b) The probability density function of the random variable X is given by **2**

$$f(x) = \begin{cases} \frac{k}{\sqrt{4-x^2}}, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Find the value of the constant k .

End of Question 7

Question 8 (6 marks) Start a new page

(a) Find the constant term in the expansion of $\left(x - \frac{2}{x^3}\right)^{20}$. **2**

(b) Given that **4**
$$(2 + x)^5(1 + 2x)^n = 32 + ax + 27600x^2 + \cdots + 2^n x^{n+5}$$

find the values of the integers a and n .

End of Question 8

Question 9 (7 marks) Start a new page

(a) Consider the parallelogram $OABC$. Let

$\mathbf{a}$ be the position vector of point A

$\mathbf{c}$ be the position vector of point C

- (i) Express $\overrightarrow{OB}$ and $\overrightarrow{CA}$ in terms of $\mathbf{a}$ and $\mathbf{c}$. **2**
- (ii) Show that if $|\overrightarrow{OB}| = |\overrightarrow{CA}|$ then $\mathbf{a} \perp \mathbf{c}$. **2**
- (iii) Show that if $\overrightarrow{OB} \perp \overrightarrow{CA}$ then $|\mathbf{a}| = |\mathbf{c}|$. **2**
- (iv) If both assumptions of (ii) and (iii) hold, describe the result geometrically. **1**

End of Question 9

Question 10 (9 marks) Start a new page

- (a) The position vectors of A , B and C referred to a point O are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively. The point N is on AB such that $AN : NB = 2 : 1$.

(i) Show that $\overrightarrow{ON} = \frac{1}{3}(\mathbf{a} + 2\mathbf{b})$. **2**

(ii) If O is the midpoint of CN , prove that $\mathbf{a} + 2\mathbf{b} + 3\mathbf{c} = \mathbf{0}$. **1**

(iii) The point M is on CB such that $CM : MB = 2 : 3$. Show that A , O and M are collinear. **2**

(iv) Hence, find the ratio $AO : OM$. **1**

(v) If the point P is such that $\overrightarrow{NP} = \overrightarrow{AM}$, show that the ratio of
area of $PNAM$: area of $PNOM = 12 : 7$ **3**

End of Question 10

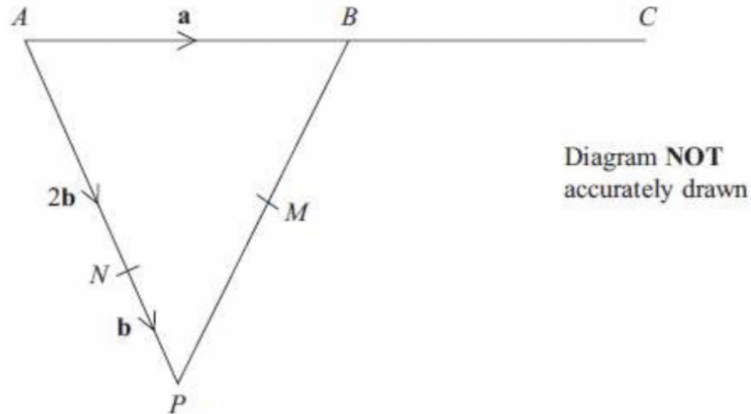
Question 11 (7 marks) Start a new page

- (a) Let X be a random variable with mean 25 and variance 4. If $P(Z < 1.5) = 0.9332$, then find $P(|X - 25| < 3)$. **3**
- (b) Let $f(x) = a(3e^{-|x|} - 1)$ for $-\ln 3 \leq x \leq \ln 3$ be a probability density function of a random variable X where a is a constant. Find the cumulative distribution function of X in terms of x and a . **4**

End of Question 11

Question 12 (7 marks) Start a new page

- (a) APB is a triangle. N is a point on AP . $\overrightarrow{AB} = \mathbf{a}$, $\overrightarrow{AN} = 2\mathbf{b}$, $\overrightarrow{NP} = \mathbf{b}$.



- (i) Find the vector $\overrightarrow{PB}$, in terms of $\mathbf{a}$ and $\mathbf{b}$. **1**
- (ii) B is the midpoint of AC . M is the midpoint of PB . Show that NMC is a straight line using vector methods. **3**
- (b) By considering the expansion of $(1+x)^n$, show that **3**

$$1 - \frac{1}{2} \binom{n}{1} + \frac{1}{3} \binom{n}{2} + \cdots + (-1)^n \frac{1}{n+1} \binom{n}{n} = \frac{1}{n+1}$$

End of Question 12

End of Examination.

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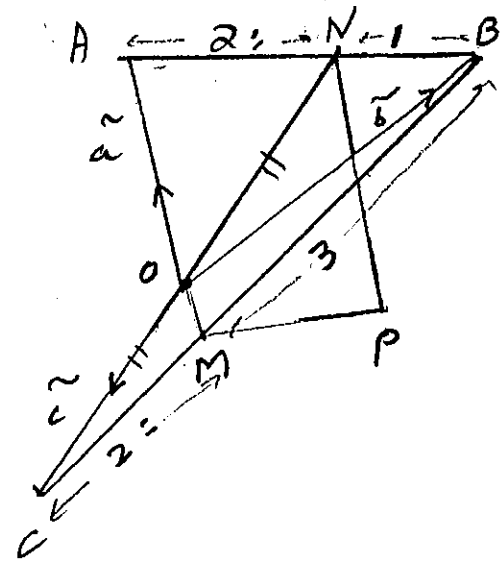
$$(i) \vec{AN} = \frac{2}{3}(\vec{b} - \vec{a})$$

$$\vec{ON} = \vec{OA} + \vec{AN}$$

$$= \vec{a} + \frac{2}{3}(\vec{b} - \vec{a})$$

$$= \frac{1}{3}[\vec{a} + 2\vec{b}]$$

$$\text{or } \vec{ON} = \vec{b} - \frac{1}{3}(\vec{b} - \vec{a})$$



$$(ii) O \text{ is midpoint of } CN$$

$$\therefore \vec{CO} = \vec{ON} = -\vec{c}$$

$$\frac{1}{3}(\vec{a} + 2\vec{b}) = -\vec{c}$$

$$\therefore \vec{a} + 2\vec{b} + 3\vec{c} = 0$$

$$(iii) \vec{AM} + \vec{MB} = \vec{AB} \therefore \vec{AM} = \vec{AB} - \vec{MB}$$

$$\vec{AM} = \vec{b} - \vec{a} - \frac{3}{5}(\vec{b} - \vec{c})$$

$$= \vec{b} - \vec{a} - \frac{3}{5}\vec{b} + \frac{3}{5}\vec{c}$$

$$= \frac{2}{5}\vec{b} + \frac{3}{5}\vec{c} - \vec{a}$$

$$= \frac{1}{5}[2\vec{b} + 3\vec{c}] - \vec{a}$$

$$= \frac{1}{5}(-\vec{a}) - \vec{a} \quad \text{from (i)}$$

$$= -\frac{6}{5}\vec{a}$$

$$\therefore \vec{AM} = -\frac{6}{5}\vec{OA}$$

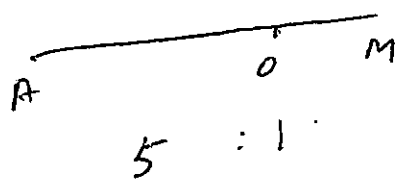
Since $\vec{AM}$ is a scalar multiple of $\vec{OA}$ and share common A

$\therefore A, O, M$ are collinear

(Alternatively prove $\vec{OM} = -\frac{1}{5}\vec{OA}$)

iv) From (iii) $|AM| = \frac{6}{5} |OA|$

P 2/2



$\therefore AO = OM = 5:1$

-5:1 get OM

v) $\vec{NP} = \vec{AM}$ given
 $\therefore PNAM$ is a parallelogram
 $|PNAM| = h |\vec{AM}|$ while h is the perpendicular from M to NP
 $= h \cdot \frac{6}{5} |AO|$

$PNOM$ is a trapezium

$|PNOM| = \frac{h}{2} [|NP| + |OM|]$
 $= \frac{h}{2} [|AM| + \frac{1}{5} |AO|]$
 $= \frac{h}{2} [\frac{6}{5} |AO| + \frac{1}{5} |AO|]$
 $= \frac{h}{2} [\frac{7}{5} |AO|]$
 $= \frac{7}{10} |AO| \cdot h$

$\frac{|PNAM|}{|PNOM|} = \frac{\frac{6}{5} h |AO|}{\frac{7}{10} h |AO|} = \frac{6}{5} \times \frac{10}{7} = \frac{12}{7}$

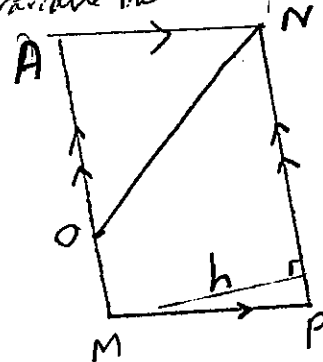
$\therefore \text{Area } PNAM : \text{Area } PNOM = 12 : 7$

Most student forgot to prove

$PNAM$ is

parallelogram
 will get 2m

max.
 Some students introduce new variable but never define it



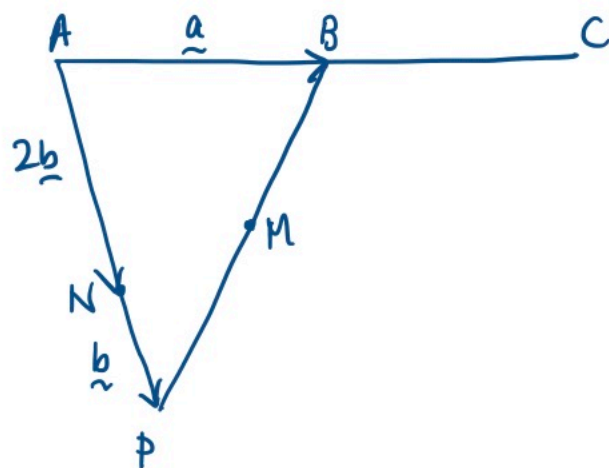
v. poorly done
 Students should
 draw diagram
 to illustrate better

MATHEMATICS EXTENSION 1 : Question 12

Suggested Solutions

Marks

Marker's Comments



$$\begin{aligned}
 \text{(i)} \quad \vec{PB} &= \vec{PA} + \vec{AB} \\
 &= -\underline{b} - 2\underline{b} + \underline{a} \\
 &= \underline{a} - 3\underline{b}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \vec{NM} &= \vec{NP} + \vec{PM} \\
 &= \vec{NP} + \frac{1}{2} \vec{PB} \\
 &= \underline{b} + \frac{1}{2} (\underline{a} - 3\underline{b}) \\
 &= \frac{1}{2} (\underline{a} - \underline{b})
 \end{aligned}$$

$$\begin{aligned}
 \vec{MC} &= \vec{MB} + \vec{BC} \\
 &= \frac{1}{2} \vec{PB} + \vec{AB} \\
 &= \frac{1}{2} (\underline{a} - 3\underline{b}) + \underline{a} \\
 &= \frac{3}{2} (\underline{a} - \underline{b}) \\
 &= 3\vec{NM}
 \end{aligned}$$

$\therefore$ PMC is a straight line.
 ($\vec{MC} \parallel \vec{NM}$ and N is a common point)

Alternative methods:

eg. Show that $\vec{NM} = \frac{1}{4} \vec{NC}$

Students need to take care with vector notation. In HSC each question will be marked independently of the rest. Could lose marks multiple times.

correct expression for $\vec{NC}$ (or equivalent)

Students must show that one vector is a scalar multiple of the other and students must state reason for collinearity

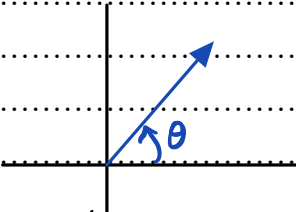
MATHEMATICS Extension 1 : Question...!!....

Suggested Solutions	Marks	Marker's Comments
a) $\sigma = 25$ $sd = 2$ $P(x - 25 < 3) = P(-3 < x - 25 < 3)$ $= P\left(-\frac{3}{2} < \frac{x - 25}{2} < \frac{3}{2}\right)$ $= 2P\left(0 < \frac{x - 25}{2} < \frac{3}{2}\right)$ $= 2 \times (0.9332 - 0.5)$ $= 0.8664$	1 1 1	correct sd and recognise the meaning of the absolute value ie. $22 < x < 28$ for recognising $\frac{x - 25}{2}$ is the z-score which is between $-\frac{3}{2}$ and $\frac{3}{2}$
b) $f(x) = a(3e^{-1x} - 1)$, $-\ln 3 \leq x \leq \ln 3$ for $-\ln 3 \leq x < 0$ $F(x) = a \int_{-\ln 3}^x (3e^t - 1) dt$ $= a [3e^t - t]_{-\ln 3}^x$ $= a(3e^x - x - 3e^{-\ln 3} - \ln 3)$ $= a(3e^x - x - 1 - \ln 3)$ $0 \leq x \leq \ln 3$ $F(x) = a \int_{-\ln 3}^0 (3e^t - 1) dt + a \int_0^x (3e^{-t} - 1) dt$ $= a[3e^t - t]_{-\ln 3}^0 + a[-3e^{-t} - t]_0^x$ $= a(3 - 0 - (3e^{-\ln 3} - (-\ln 3)))$ $+ a(-3e^{-x} - x - (-3e^0))$	1	① need to recognise the two functions and integrate correctly.

MATHEMATICS Extension 1 : Question...!!....

Suggested Solutions	Marks	Marker's Comments
$= a(3 - 1 - \ln 3)$ $+ a(-3e^x - x + 3)$ $= a(-3e^{-x} - x + 5 - \ln 3)$ $\therefore F(x) = \begin{cases} a(3e^x - x - 1 - \ln 3), & -\ln 3 \leq x < 0 \\ a(-3e^{-x} - x + 5 - \ln 3), & 0 \leq x \leq \ln 3 \end{cases}$ Note: - cannot integrate absolute values, if students have done so, a mark of 0 was awarded. - There was no mark allocated for finding 'a' $a = \frac{1}{4 - 2\ln 3}$	1 1	Also correct for $0 \leq x \leq \ln 3$: $a(-3e^{-x} - x)$ $+ 3a + \frac{1}{2}$ OR $a(-3e^{-x} - x)$ $+ a(1 + \ln 3) + 1$

MATHEMATICS EXT 1: Question 7

Suggested Solutions	Marks	Marker's Comments
ai, displacement vector = $(2\hat{i} + 10\hat{j}) - (9\hat{i} - 6\hat{j})$ $= 12\hat{i} + 16\hat{j}$ _____ ① speed = velocity $= \left \frac{\text{displacement}}{\text{time}} \right$ $= \left \frac{12\hat{i} + 16\hat{j}}{4} \right$ $= 3\hat{i} + 4\hat{j}$ $= 5 \text{ km/h}$ _____ ① ii, We want the direction of $12\hat{i} + 16\hat{j}$  $\theta = \tan^{-1}\left(\frac{16}{12}\right)$ $= 53^{\circ}8'$ $\therefore$ direction of S is 037°T or $N37^{\circ}\text{E}$ iii, In 4 hours, the displacement is $12\hat{i} + 16\hat{j}$ $\therefore$ In 1 hour, the displacement is $3\hat{i} + 4\hat{j}$ _____ ① $\therefore$ In t hours, the displacement is $t(3\hat{i} + 4\hat{j})$		Many lost the mark from not having bearing in correct form.

MATHEMATICS EXT 1: Question 7

Suggested Solutions

Marks

Marker's Comments

So the position vector of S is

$$\underline{s} = (9\hat{i} - 6\hat{j}) + t(3\hat{i} + 4\hat{j})$$

$$= (3t+9)\hat{i} + (4t-6)\hat{j} \quad \text{--- ①}$$

Comments:

* Students who subbed $t=0$ & 4 into $\underline{s} = 21\hat{i} + 10\hat{j}$ receive only one mark, unless they explicitly state that "since the equation satisfied two known points, it must be the equation of the line."

* The following solution only received one mark as well:

$$\underline{s} = \text{starting point} + t(\text{direction})$$

$$= \begin{pmatrix} 9 \\ -6 \end{pmatrix} + t \begin{pmatrix} 12 \\ 16 \end{pmatrix}, \quad t \in \mathbb{R}$$

$$= \begin{pmatrix} 9 \\ -6 \end{pmatrix} + 4t \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \quad t \in \mathbb{R}$$

$$= \begin{pmatrix} 9 \\ -6 \end{pmatrix} + t \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \quad t \in \mathbb{R}$$

$$= (3t+9)\hat{i} + (4t-6)\hat{j}$$

↳ t is NOT an arbitrary constant in this scenario. It represents time, so your direction vector must be the displacement for one change in t (i.e. one hour)

MATHEMATICS EXT 1: Question 7

Suggested Solutions	Marks	Marker's Comments
b, Since $f(x)$ is a PDF: $\int_{-\infty}^{\infty} f(x) dx = 1$ $\therefore \int_0^1 \frac{k}{\sqrt{4-x^2}} dx = 1$ $\left[k \sin^{-1}\left(\frac{x}{2}\right) \right]_0^1 = 1$ $k\left(\frac{\pi}{6} - 0\right) = 1$ $k = \frac{6}{\pi}$	① ①	Too many errors or "overkill" methods for this integral. It's a STANDARD FORM! Know your reference sheet & don't do unnecessary trig subs.

Question 6:

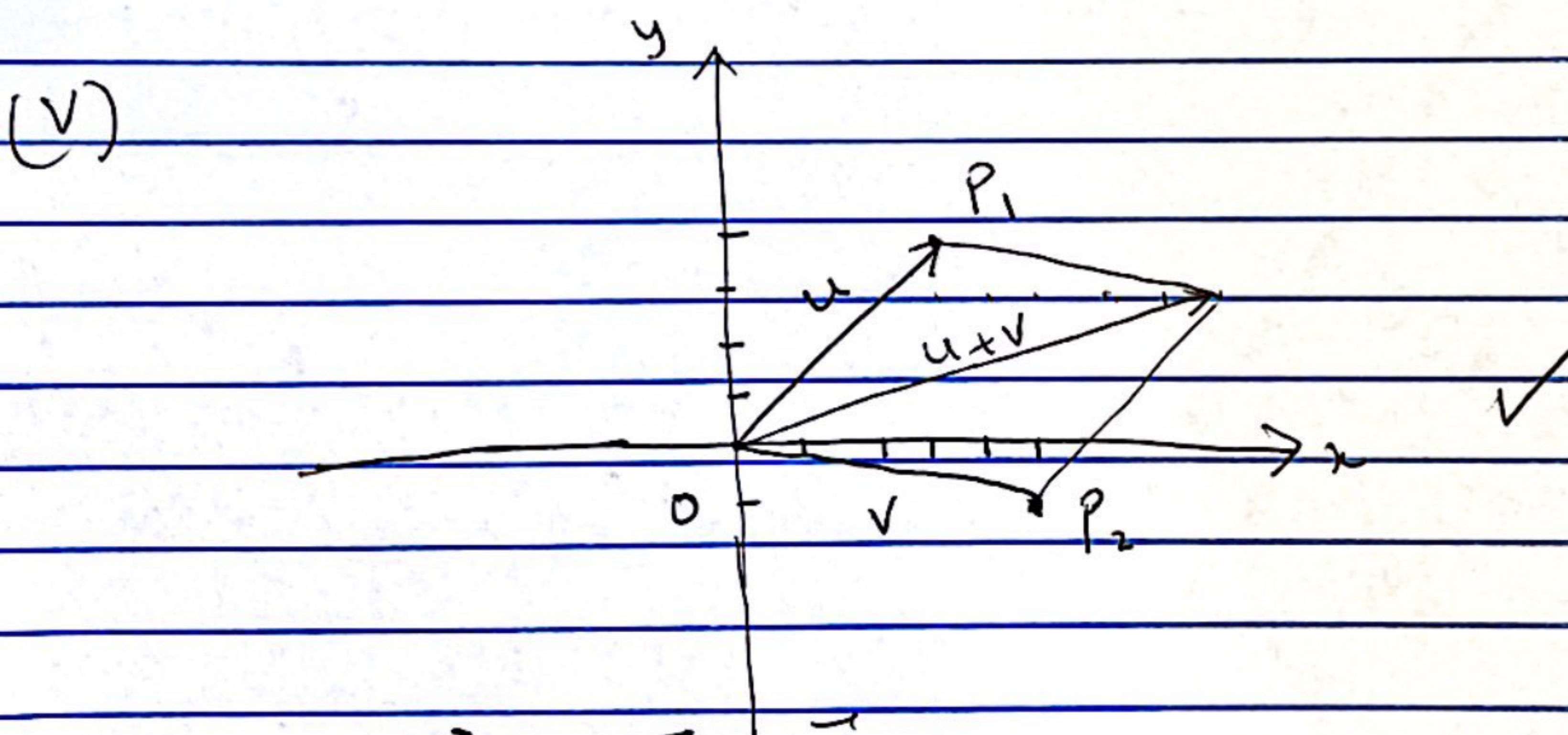
$$P_1(3, 4), P_2(5, -1)$$

(i) $\vec{OP}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ ✓

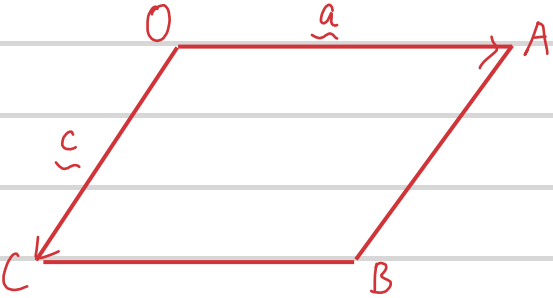
(ii) $\vec{P_1P_2} = \vec{OP_2} - \vec{OP_1}$
 $= 5\vec{i} - \vec{j} - (3\vec{i} + 4\vec{j})$ ✓
 $= 2\vec{i} - 5\vec{j}$ ✓

(iii) $|\vec{P_1P_2}| = \sqrt{2^2 + (-5)^2} = \sqrt{29}$ ✓

(iv) $u = \vec{OP_1}, v = \vec{OP_2}$
 $\vec{P_1P_2} = \vec{OP_2} - \vec{OP_1} = v - u$ ✓



(vi) $\vec{AP_2} = 2\vec{i} - 3\vec{j}$
 $\vec{OP_2} - \vec{OA} = 2\vec{i} - 3\vec{j}$
 $\vec{AO} = 2\vec{i} - 3\vec{j} - \vec{OP_2}$
 $= 2\vec{i} - 3\vec{j} - (5\vec{i} - \vec{j})$
 $= -3\vec{i} - 2\vec{j}$ ✓

Suggested Solutions	Marks	Marker's Comments
a) 		
i) $\vec{OB} = \underline{a} + \underline{c}$	1	
$\vec{CA} = \underline{a} - \underline{c}$	1	
ii) If $\vec{OB} = \vec{CA}$, then $\vec{OB} ^2 = \vec{CA} ^2$ $\vec{OB} \cdot \vec{OB} = \vec{CA} \cdot \vec{CA}$ $(\underline{a} + \underline{c}) \cdot (\underline{a} + \underline{c}) = (\underline{a} - \underline{c}) \cdot (\underline{a} - \underline{c})$ $\underline{a} ^2 + 2\underline{a} \cdot \underline{c} + \underline{c} ^2 = \underline{a} ^2 - 2\underline{a} \cdot \underline{c} + \underline{c} ^2$ $2\underline{a} \cdot \underline{c} = -2\underline{a} \cdot \underline{c}$ $4\underline{a} \cdot \underline{c} = 0$ $\therefore \underline{a} \cdot \underline{c} = 0$ $\therefore \underline{a} \perp \underline{c}$	1	
iii) If $\vec{OB} \perp \vec{CA}$, $\vec{OB} \cdot \vec{CA} = 0$ $(\underline{a} + \underline{c}) \cdot (\underline{a} - \underline{c}) = 0$ $\underline{a} \cdot \underline{a} - \underline{c} \cdot \underline{c} = 0$ $\underline{a} ^2 - \underline{c} ^2 = 0$ $\underline{a} ^2 = \underline{c} ^2$ $\underline{a} = \underline{c}$ (both positive)	1	
iv) OACB is a square (parallelogram with one pair of adjacent perpendicular sides)	1	

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

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[illegible]

[illegible]

Year 12 Extension 1: Q8		
Suggested Solutions	Marks Awarded	Marker's Comments
(a) General term of $(x - 2x^{-3})^{20}$ is $\binom{20}{k} x^{20-k} (-2x^{-3})^k = \binom{20}{k} (-2)^k x^{20-4k}$ Constant term occurs when $20 - 4k = 0 \rightarrow k = 5$: $\binom{20}{5} (-2)^5 = -496128$ (b) $(2 + x)^5 (1 + 2x)^n = \left(\sum_{k=0}^5 \binom{5}{k} 2^k x^{5-k} \right) \left(\sum_{j=0}^n \binom{n}{j} 2^j x^j \right)$ $= \sum_{k=0}^5 \sum_{j=0}^n \binom{5}{k} \binom{n}{j} 2^{k+j} x^{5-k+j}$ We first need to extract the coefficient of x^2. The simplest way to do so is to take expansions to x^2 in each binomial and multiply judiciously: $(32 + 80x + 80x^2 + \dots)(1 + 2nx + 2n(n-1)x^2 + \dots)$ $= 32 + (64n + 80)x + (64n(n-1) + 160n + 80)x^2 + \dots$ We have then $64n(n-1) + 160n + 80 = 27600$ so $64n^2 + 96n - 27520 = 0$ which gives, using quadratic formula, $n = 20$ as the only positive, integral solution. Hence, the coefficient of x is $a = 64n + 80$ $= 64 \cdot 20 + 80$ $= 1360$	1 First mark for appropriate working leading to a general term. 1 Final mark for correct result. 1 First mark for work leading to expansion. 1 Next mark for acquiring correct equation in n. 1 Mark for extracting correct n. 1 Final mark for correct value of a.	Many errors made, primarily since candidates did not multiply expansions correctly. ECF marks not necessarily awarded if errors simplified the problem.